Mohammed Esamaldin Mahjoub Mohammed

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Research interests: internals-based misalignment monitors, mechanistic interpretability, agent evaluation, AI safety for low-resource communities

Summary

Chevening Scholar and AI research engineer from Sudan, based in London. I work on cheap monitors that read model internals to detect misalignment before it shows in behaviour, built for open-weight models so that low-resource communities can oversee the AI systems they deploy. I am currently a BASE Fellow (AI Catastrophic Risk track) studying whether optimisation geometry predicts emergent misalignment, alongside independent research on attributing goal drift in LLM agents from internal representations. I also helped start AI Safety Arabia, which explains AI safety risks to Arabic-speaking audiences.

Research

Can Optimisation Geometry Predict Emergent Misalignment? *Oct 2026 – present*

BASE Fellowship, AI Catastrophic Risk track (selected from 1,200+ applicants). Mentor: Sultan Hassan. Team project, in progress

- Reproducing emergent misalignment from narrow fine-tuning on 1–8B open-weight models, measuring alignment against coherence with an LLM judge.
- Tracking loss-landscape geometry across fine-tuning checkpoints on 8–12B models to test whether it signals misalignment before behavioural evaluations do.

Forgotten, Inferred, or Substituted: Attributing Goal Drift in LLM Agents from Internal Representations *2026 – present*

Independent research, in progress, [Proposal]

- Designing a testbed that produces three causes of goal drift (interference, conditioning on an inherited drifted trajectory, and backdoored goal substitution) in one agent environment, matched on trajectory length, step count, tool-call volume, task success, and external suspicion score.
- Testing whether internal representations of Llama-3.1-8B and Qwen3-14B separate the causes where behaviour cannot, against TACT, Fatigue Index, and step-count baselines, with cross-model and cross-organism generalisation tests and a pre-stated kill condition for each research question.

ZANOBIA: A Socratic Framework for Bidirectional Detection and Mitigation of Hallucinations in Multimodal LLMs *2025*

Mohammed, M.E.M. MSc thesis (Distinction). [Paper] [Code] [Write-up]

- Designed and built a framework that detects and mitigates hallucinations from both the user side and the model side: it clarifies ambiguous prompts with Socratic questioning and verifies model outputs against external knowledge and visual evidence, with no retraining.
- Reduced the hallucination rate on MMHal-Bench from 0.76 to 0.64 (16% lower) with LLaVA-7B.
- Targeted user-induced hallucinations, to which prior work (Wu et al.) attributes roughly 46% of factual errors, by treating user ambiguity as a distributional shift.

Defensive AI Governance for Fragile States: A Sovereignty-First Framework *2024*

Mohammed, M.E.M. [Draft]

- Proposed a framework for AI deployment in states without formal regulatory infrastructure, using Sudan as the primary case study.
- Synthesised the EU AI Act, NIST, and Singapore frameworks into seven principles enforced through soft-law levers (procurement rules, audits, transparency labels).

AI Safety Field-Building

Founding member, AI Safety Arabia (AISA) *2026 – present*

- Helped start an Arabic-language AI safety education platform at the Translating AI Safety hackathon run by LIaiSE (London Interface for AI Safety); 4 videos published on misalignment and loss-of-control risks.

Technical Workshop Lead, Civil Registry Authority, Sudan *2022*

- Designed and delivered a curriculum on data privacy and AI governance for 30+ government engineers and policymakers from Sudan's three largest civil-registration centres.

AI Safety Training

ML4Good Technical Bootcamp, ML4Good [Certificate] *Sep 2026*

- 8-day in-person intensive covering hands-on deep learning implementation, AI safety research, and technical AI governance.

Technical AI Safety, BlueDot Impact [Certificate] *Jun 2026 – Jul 2026*

- 30-hour cohort course covering current safety techniques and their gaps (alignment and RLHF, interpretability, evaluations and red-teaming, AI control, scalable oversight), threat modelling via kill-chain analysis, and a personal action plan.

Introduction to Cooperative AI, Cooperative AI Foundation [Certificate] *Jul 2026 – Aug 2026*

- 8-week cohort on multi-agent coordination, game theory, and infrastructure for beneficial AI.

Education

MSc Artificial Intelligence (Distinction), Queen Mary University of London *Sep 2024 – Sep 2025*

- Chevening Scholar. Thesis: ZANOBIA (see Research).
- Modules: Deep Learning, Machine Learning, NLP, Information Retrieval, Applied Statistics, Ethics, Regulation and Law in Advanced Digital Information Processing.

BSc Electrical Engineering, Sudan University of Science and Technology *Oct 2014 – Dec 2020*

- Thesis: Application of Computer Vision in Precision Agriculture, Aerial Mapping via Quadcopter.
- Modules: Signal Processing, Artificial Neural Networks, Linear Algebra, Probability and Applied Statistics, Differential Equations.

Professional Experience

AI Research Engineer, Q parts *Jul 2023 – Aug 2024*

- Built a production parts-identification conversational AI that won Digital Product of the Year (Automechanika Dubai 2024): a LangChain pipeline on GPT-4o with retrieval over BERT embeddings and TF-IDF cosine similarity.
- Owned the retrieval pipeline end to end, from ingestion and embedding to similarity search and response grounding, tuning retrieval quality against relevance in production.
- Built reproducible pipelines with Docker and Airflow, with strict version control and experiment tracking for large-scale data collection.

Data Scientist, Morouj *Dec 2022 – Feb 2023*

- Designed and validated a turnover-prediction pipeline on 250k+ HR records, improving baseline accuracy by 15% through feature engineering, with A/B testing and experiment-tracking dashboards to catch failure modes in deployment.

ML Engineer, TRX *Nov 2020 – Jul 2021*

- Developed reproducible ETL pipelines and TensorFlow-based automation for behavioural analytics across event streams for 110k+ users, with input validation and retry logic for data integrity.

Funding & Awards

- **BASE Fellowship**, AI Catastrophic Risk track (2026). Selected from 1,200+ applicants.
- **Chevening Scholarship**, UK Foreign, Commonwealth & Development Office (2024). Selected from the top 2% of global applicants.
- **EAGxOxford** (2026). Fully funded attendee.
- **Digital Product of the Year**, Automechanika Dubai (2024), for the Q parts system above.
- **Technology Incubation Program**, African Development Bank (2021). Selected from 600+ applicants across Africa to incubate SudaAgro, an AI precision-agriculture startup built on my BSc thesis.
- **Raid Incubation Program**, Sudan Foundation (2019). Competitive 18-month acceleration grant.
- **National Academic Merit Scholarship**, Government of Sudan (2014). Full undergraduate tuition.

Skills

- **Interpretability & evaluation:** TransformerLens, activation patching, LLM-as-judge evaluation, hallucination benchmarking (MMHal-Bench, MME), experiment design.
- **Research engineering:** PyTorch, Hugging Face Transformers, TRL, LangChain, LangGraph, Pydantic, OpenAI API.
- **Systems:** Python (advanced), Docker, Airflow, Git, CI/CD, SQL, AWS, FastAPI, PostgreSQL, Redis.
- **Mathematics:** linear algebra, probability theory, information theory, signal processing.
- **Languages:** Arabic (native), English (fluent).